

- Mesh Networking: Auto-backhaul links between APs
- Airtime Fairness: Prioritizing faster clients to improve performance
- Rogue AP Detection

Omada Wi-Fi 6 Access Points

TP-Link's Omada Wi-Fi 6 Access Points leverage the latest 802.11ax technology to deliver high-speed, efficient wireless connectivity suitable for various environments

Indoor Access Points

- **EAP670 (AX5400):** Offers combined speeds up to 5400 Mbps (4804 Mbps on 5 GHz and 574 Mbps on 2.4 GHz). Features a 2.5G Ethernet port and supports 160 MHz channels for enhanced throughput
- **EAP660 HD (AX3600):** Designed for high-density environments, it provides up to 3550 Mbps combined speeds with 8 spatial streams, ensuring robust performance in crowded areas.
- **EAP650 (AX3000):** Delivers up to 2976 Mbps combined speeds, featuring 160 MHz channel support and centralized cloud management via Omada SDN.
- **EAP620 HD (AX1800):** Ideal for high-density scenarios, offering up to 1775 Mbps combined speeds with 4 spatial streams, enhancing multi-user throughput

Outdoor Access Points

- **EAP610-Outdoor (AX1800):** Provides up to 1.8 Gbps speeds, suitable for outdoor deployments like cafés and amusement parks, ensuring reliable connectivity in various weather conditions.

Omada Layer 3 Managed Switches

Omada's L3 Managed Switches offer advanced routing capabilities, high-speed connectivity, and centralized management, making them suitable for enterprise networks.

- **Omada 24-Port Gigabit Stackable L3 Managed Switch:** Features 24× Gigabit RJ45 ports and 4× 10 Gbps SFP+ slots, with a 128 Gbps switching capacity. Supports physical stacking up to 8 units and includes L3 features like RIP, OSPF, and DHCP Server/Relay.
- **TP-Link Omada 48-Port L3 Managed PoE+ Ethernet Switch 1440W:** Offers 48× Gigabit RJ45 PoE+ ports and 6× 10 Gbps SFP+ slots, with a substantial 1440W PoE power budget. Supports L3 routing protocols and physical stacking for enhanced scalability.
- **Omada 26-Port 10G Stackable L3 Managed Aggregation Switch:** Equipped with 26× 10G SFP+ slots and 6× 25G SFP28 slots, providing an 820 Gbps switching capacity. Ideal for aggregation layers in large networks.

Omada VPN Load Balancing Routers

- TP-Link's Omada VPN Routers ensure secure, reliable internet

connectivity with features like multi-WAN load balancing and centralized management.

- **TP-Link ER605 SafeStream Gigabit Multi-WAN VPN Router:** Supports up to 3 WAN Ethernet ports and 1 USB WAN for mobile broadband. Features load balancing to optimize bandwidth usage and integrates seamlessly with Omada SDN for centralized management.
- **TP-Link ER7212PC Omada Gigabit VPN Router:** Offers 6× Gigabit ports (1× SFP, 5× RJ45), supports load balancing on up to 5 WAN ports, and includes a USB 3.0 port for LTE WAN backup. Provides various VPN protocols and centralized cloud management.
- **ER8411 Omada VPN Gateway with 10G Ports:** Features up to 10 WAN ports (SFP+, SFP, and RJ45) and one USB WAN for mobile broadband. Supports multi-WAN load balancing and integrates with Omada SDN for streamlined network management.

Common Utilities

- **Omada Discovery Utility (for Windows):** Finds and adopts devices on the LAN.
- **Ping and Traceroute:** Verify basic connectivity and routing.
- **Wireless Scanner:** Detect interference and overlapping channels.
- **Syslog / SNMP:** Monitor logs and alerts.

Logs & Monitoring

- View Logs under Insight > Logs.
- Set up email alerts for critical events (firmware upgrade, AP down, WAN failover).
- Enable Syslog server for detailed historical logs.

Pro-Tips

- Always match firmware versions with Omada Controller version compatibility chart.
- Use Tag-Based VLANs for multi-tenant or guest Wi-Fi networks.
- Schedule regular configuration backups from the Controller.

Education Sector (Schools, Colleges, Universities)

Challenges:

- ✓ High user density (students, staff, guests)
- ✓ Need for secure, segmented networks (Admin, Student, Guest)
- ✓ Campus-wide coverage and mobility

Omada SDN Benefits:

- **Centralized Management:** Control all APs, switches, and routers from one interface, whether on-site or remotely.
- **Seamless Roaming:** Students and faculty enjoy uninterrupted Wi-Fi moving between classrooms and buildings.
- **Role-Based VLANs:** Securely segment academic, administrative, and guest traffic.